DATA3 NF270 — Choosing B(c), the transport regime, and whether experiments inform each other

Single-salt diafiltration: NaCl · $CaCl_2$ · $LaCl_3$ | per-response AIC · Peclet regime · multi-experiment pooling

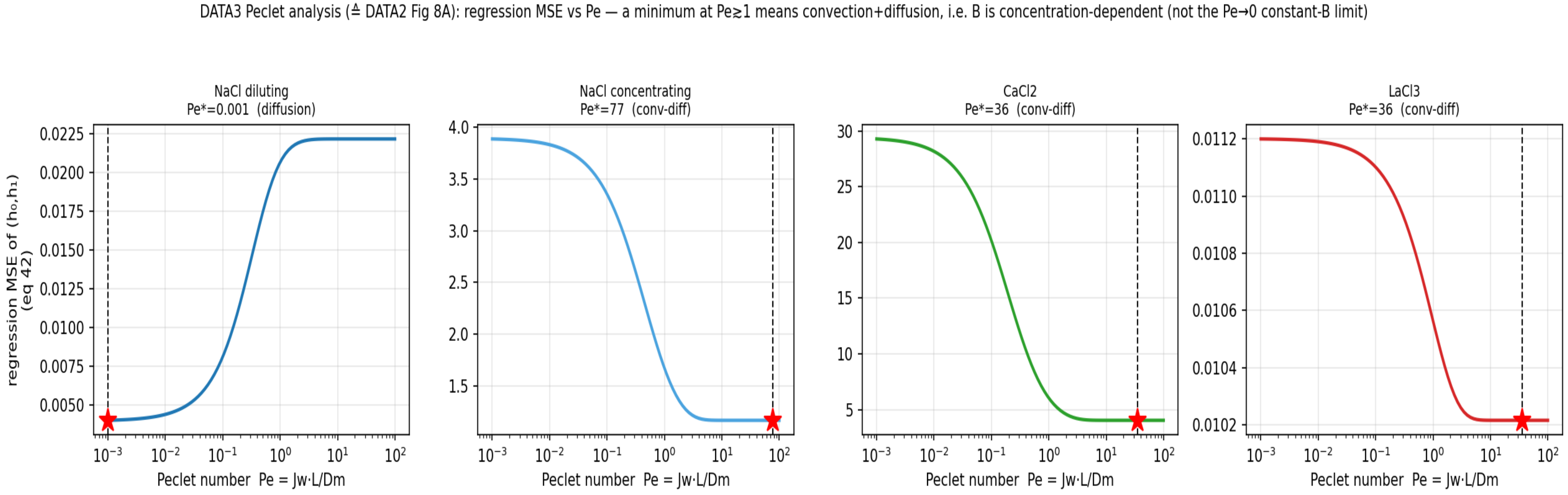
Generated from the session analyses — auto-updates as the contour campaign completes

What sets the solute permeability B, and does it depend on concentration?

- B is the membrane solute permeability: $J_s = Jw \cdot B \cdot (c\_in - c\_H)$.
- DATA1/DATA2 transport theory: $B = D_m \cdot H/L$ — a partition (H) and a diffusion length.
- Whether B is constant or concentration-dependent is set by the Peclet number $Pe = Jw \cdot L/D_m$:
  - – $Pe \to 0$ (diffusion-dominated): $B = D_m H/L$ = CONSTANT
  - – $Pe \sim 1+$ (convection + diffusion): $B = Jw \cdot \Sigma\, \beta_i\, c^i$ (concentration-dependent Taylor form)
- Under a constant-B fit, σ and B and L_p are coupled and σ rails — the identifiability problem we solve here.

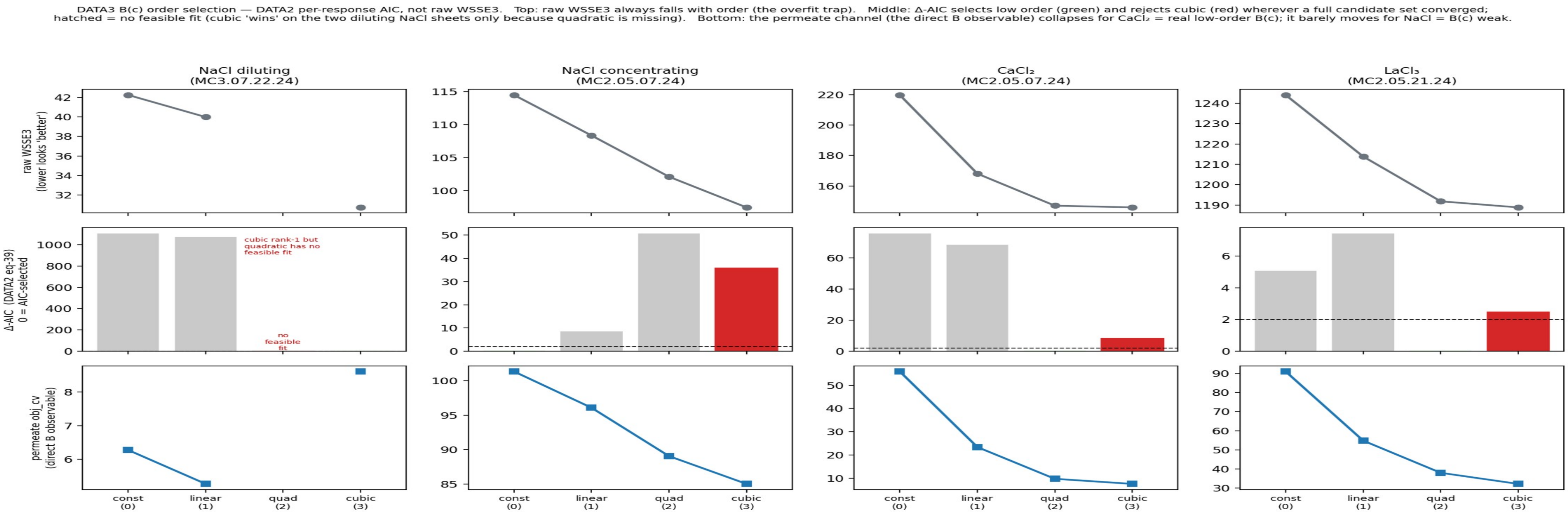
→ Three threads: which B(c) form (AIC), why it depends on c (Peclet), and can pooling experiments fix identifiability.

Regression MSE vs Pe (≙ DATA2 Fig 8): which regime each salt is in



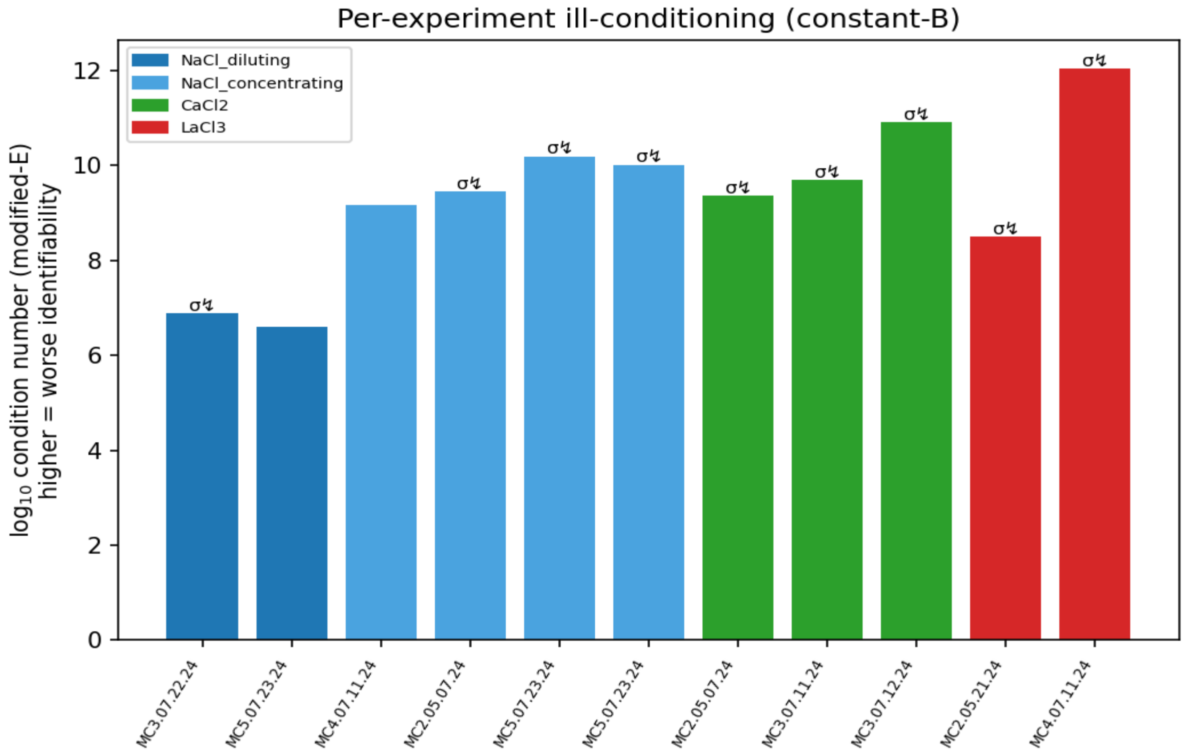
→ Diluting NaCl → Pe→0 (diffusion, B≈const). $CaCl_2$ / concentrating-NaCl / $LaCl_3$ → Pe≫1 (convection+diffusion, B(c)). The regime direction is the robust signal.

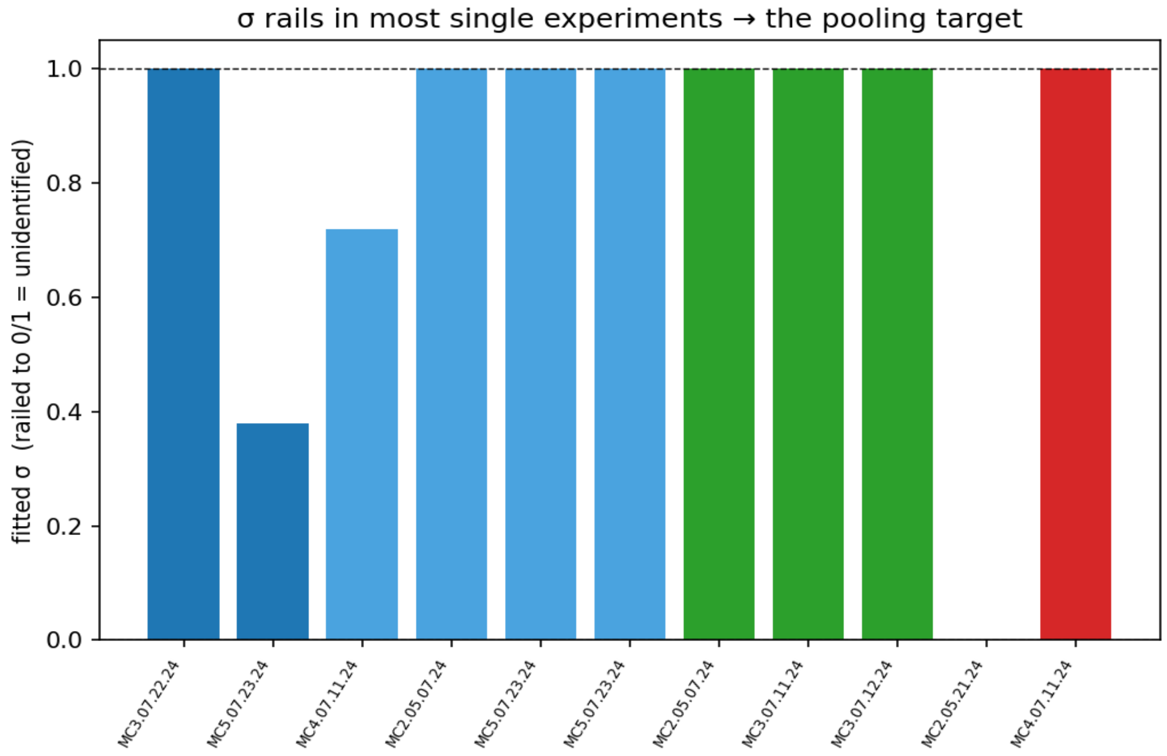
Order chosen by the per-channel AIC (eq 39), not raw WSSE3



→ $CaCl_2$ → quadratic/linear, cubic REJECTED (Δ=8–25). NaCl cubic 'wins' are mass-channel/convergence artifacts. Use the per-response AIC, not the pooled JSON AIC.

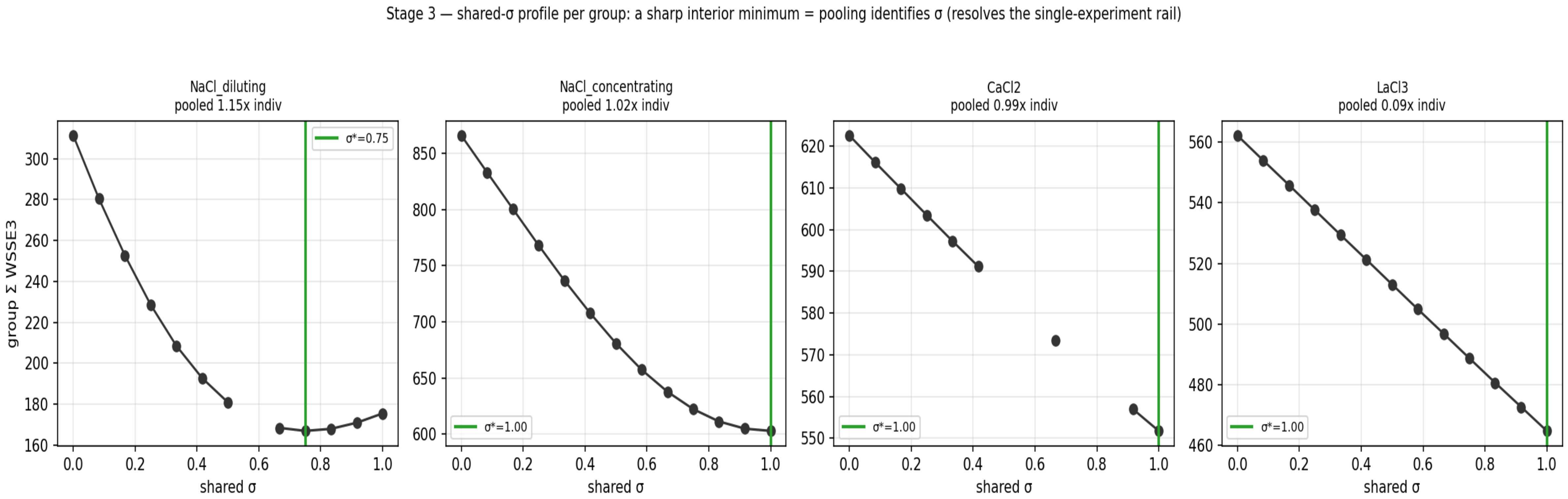
Per-experiment σ rails; the σ–B–L_p landscape is ill-conditioned





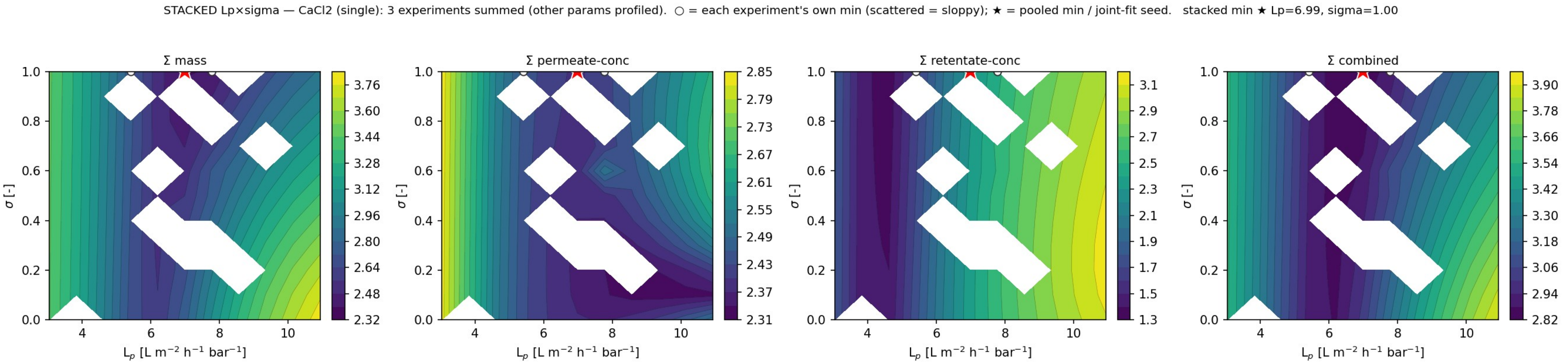
→ σ rails in 9/11 single experiments; condition numbers 10^6–10^{12}. A single experiment cannot pin σ.

One shared σ per salt-regime group costs ~nothing — and comes off the rail



→ Cost ≈1× (sharing σ is nearly free → σ is a salt property). Diluting-NaCl identifies σ*=0.75, OFF the individual rails (1.0/0.38).

Summing the objective surfaces = summing Fisher information → a sharper pooled minimum

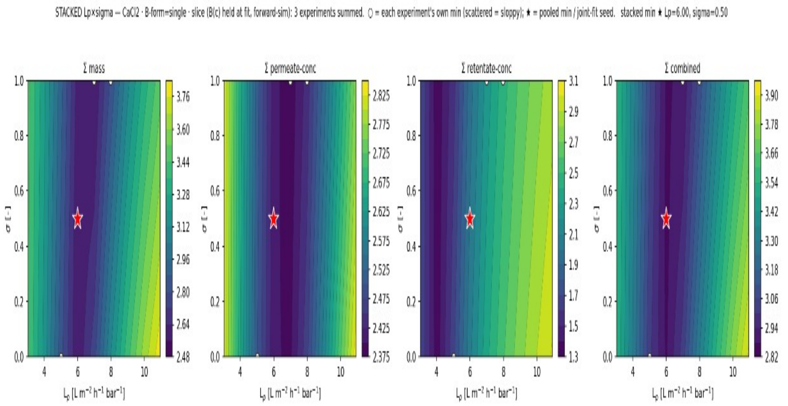


Per channel across a group: Σ mass → L_p, Σ permeate-conc → B, Σ retentate-conc → σ. White ○ = each experiment's scattered own minimum; red ★ = the pooled minimum (joint-fit seed). The transport model is evaluated at every node (slice = forward-sim, legacy calc_contour_2d; profile = re-solve B(c)).

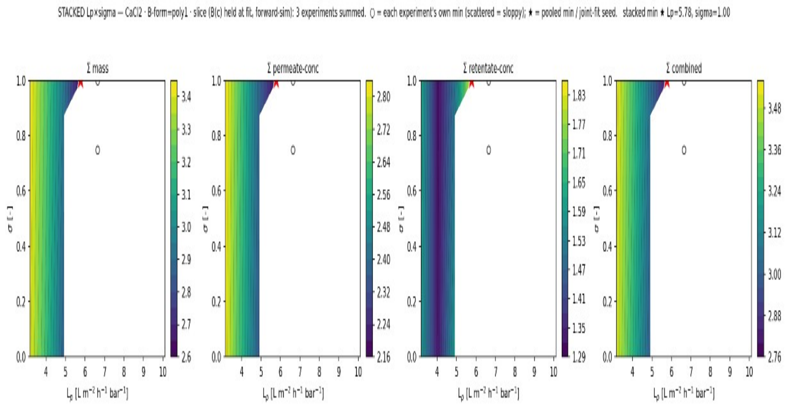
→ Where individuals are sloppy, the summed surface localizes — the visual proof that pooling improves identifiability.

Each concentration-function of B reshapes the σ/L p identifiability ($CaCl_2$)

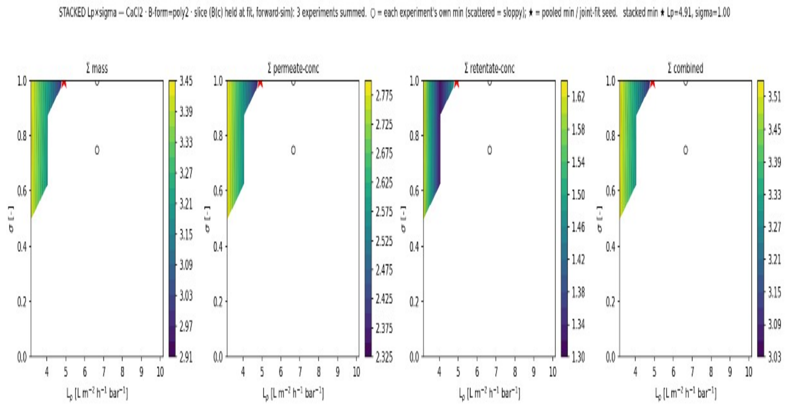
constant



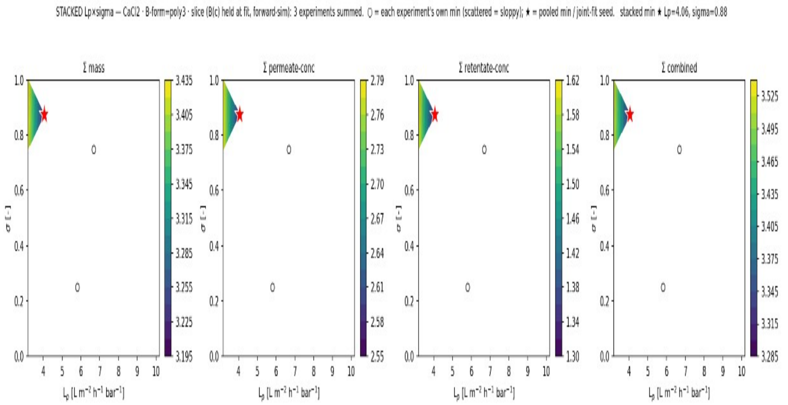
linear



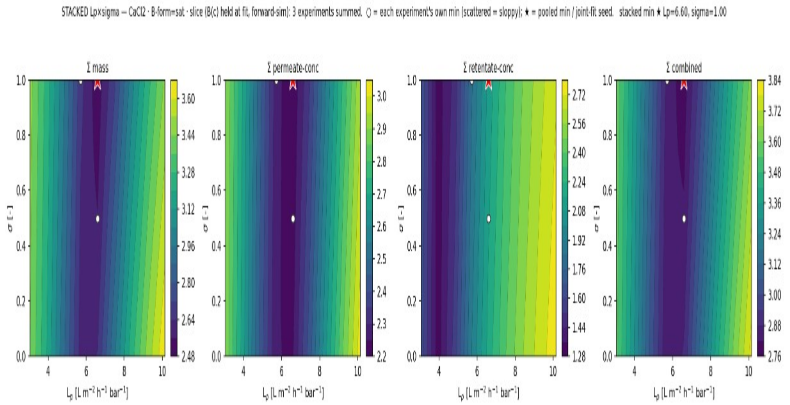
quadratic



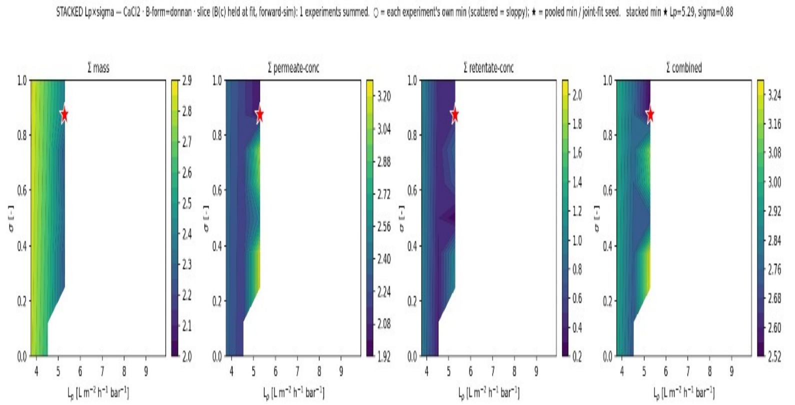
cubic



saturating



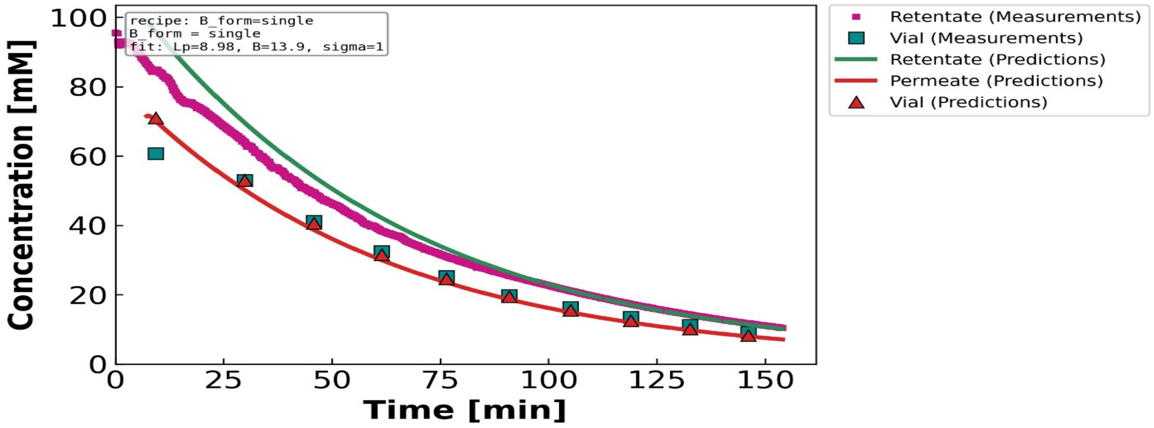
Donnan



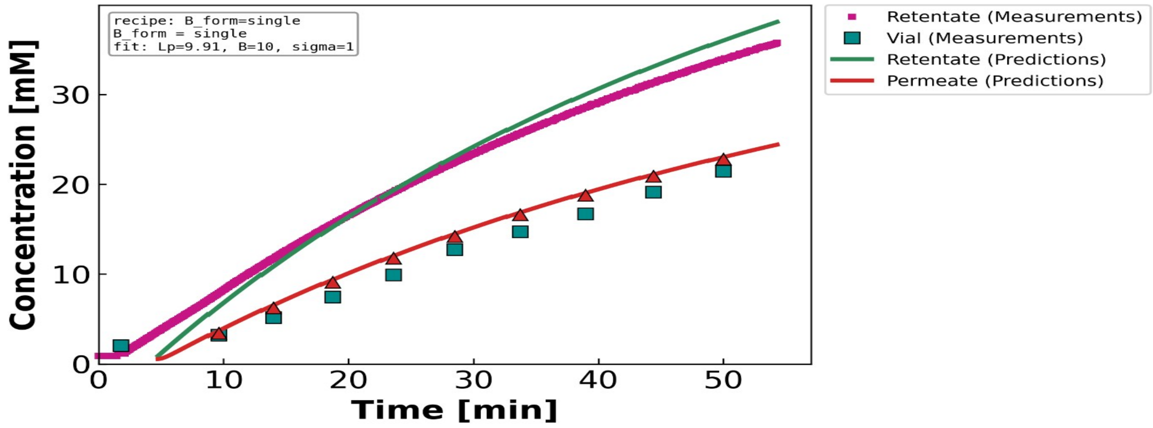
→ Comparing forms: constant rails σ; the concentration-dependent forms relax the σ/L p trade-off differently.

The first B-form (constant B) at the optimized parameters

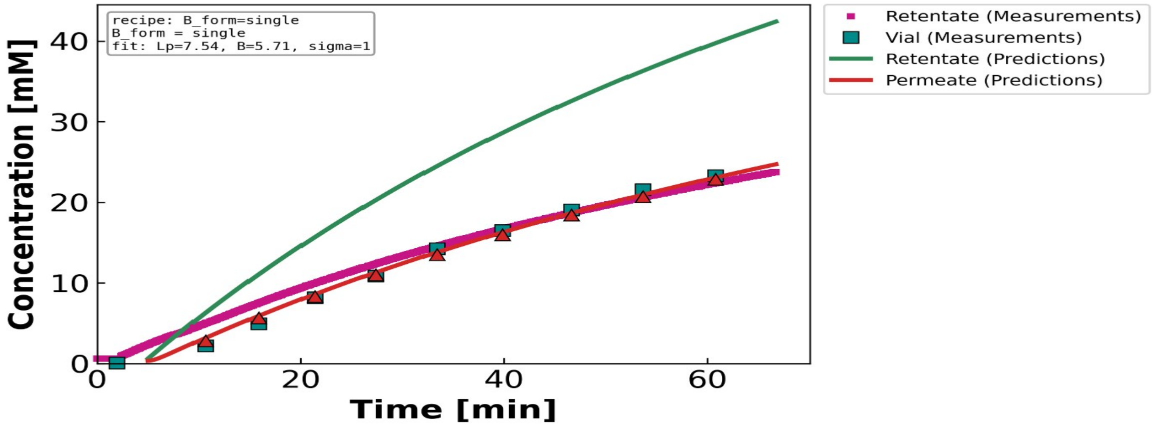
NaCl diluting



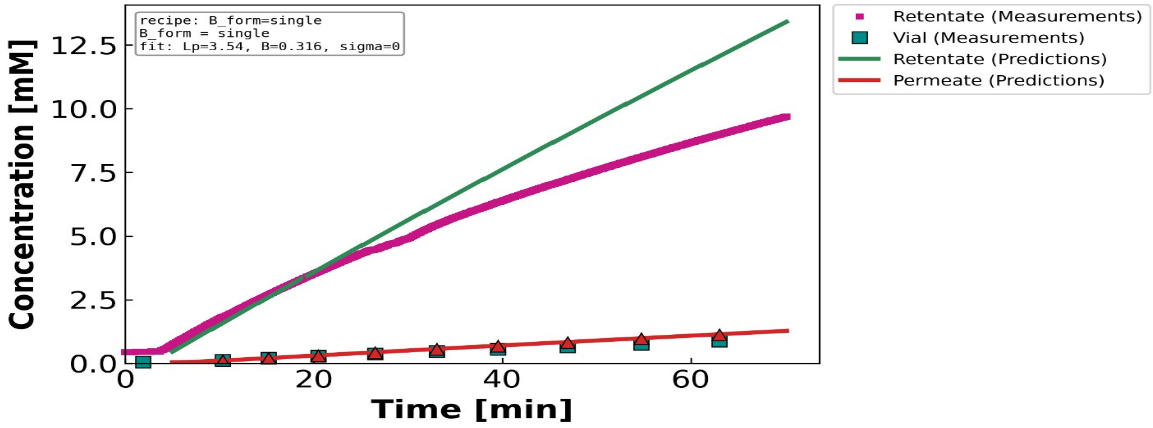
NaCl concentrating



CaCl2



LaCl3



→ Constant-B tracks NaCl well (flat B); for $CaCl_2$/$LaCl_3$ the constant-B residual motivates B(c).

1. B is constant when diffusion dominates (Pe→0) and concentration-dependent when convection matters (Pe≫1). The Peclet analysis separates the salts: diluting-NaCl diffusion-limited; $CaCl_2$/$LaCl_3$/conc-NaCl convection.

2. Among concentration-functions of B, the DATA2 per-response AIC selects LOW order — quadratic/linear for $CaCl_2$, cubic rejected. Raw WSSE3 over-selects; the per-channel likelihood is the honest criterion.

3. A single experiment cannot identify σ (it rails; cond 10^6–10^{12}). Pooling experiments of the same salt+regime shares σ at ~zero cost and resolves the rail (diluting-NaCl $\sigma^*=0.75$). Stacked objective landscapes show the experiments informing each other: the summed surface has a sharper minimum than any individual.

4. Same NF270 membrane across DATA3 → L_p is consistent; the within-group scatter was identifiability noise that pooling removes. The stacked minimum is the warm-start for the joint multi-experiment fit.